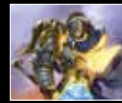


NASA craft crash lands

NASA lets the Lunar Atmosphere and Dust Environment Explorer slam into the moon after it runs out of fuel <http://dgit.in/LADEECrash>



34 down – 162 to go

Read about the game that became the most-downloaded iPad app in 34 different countries a day after it launched <http://dgit.in/hrths34>

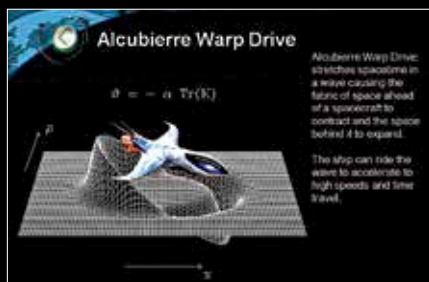
would allow the spacecraft to travel at nearly 10 times the speed of light - making the journey to the Moon in one twelfth of a second, to Pluto in 33 minutes and to Alpha Centauri in 159 days.

Of course this technology would require the harnessing of specific exotic particles and have a host of side-effects, from complete lack of steering control to a sonic boom shockwave that would destroy anything in front of the spacecraft at its destination. But despite the potential impossibility of a technological device making the Alcubierre drive a reality, it is clear that the universe as we understand it allows its existence. And that is sufficient for us to figure out the rest.

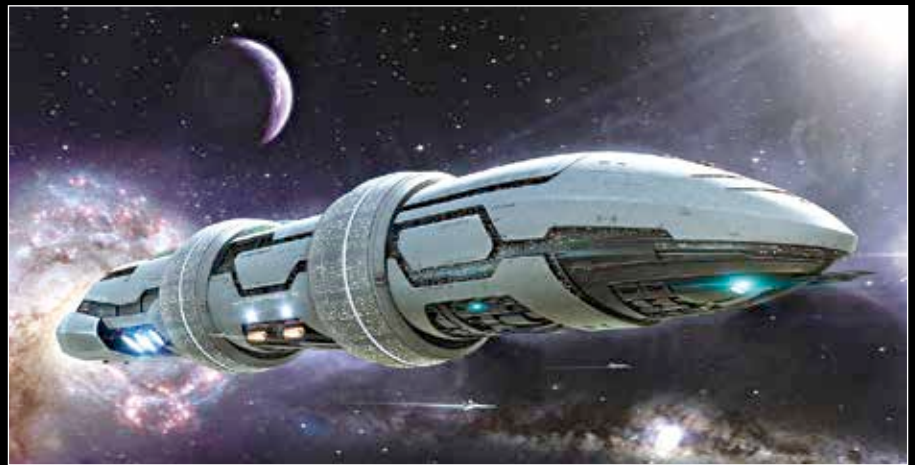
Interstellar Ark For the Future

Another major "to-do" item for mankind is to plan for another home planet. This "single-point failure" is within the very nature of humanity and the risks are plenty – natural resource exhaustion, climate change and the next Extinction Level Event like an asteroid strike.

This reality was further felt when in December 2004 a near-Earth asteroid was discovered called "99942 Apophis" that had a frightening chance of striking the Earth in 2036. This asteroid was initially predicted to crash in the Pacific Ocean causing devastating tsunamis and floods.



With enough exotic matter scientists already have an idea of how a Alcubierre Drive ship could be designed.



Imagine an interstellar ark as a cross between a cruise ship and a labour camp.

Fortunately, further research concluded that Apophis would miss the Earth completely but it did bring the problem to the forefront with Russia at one point preparing an Armageddon-like interception program.

The next level of interstellar spacecraft could very well be built with the intention of launching generational ships that travel across the galaxy to colonise Earth-like planets decades or centuries later. We have already begun to catalogue such planets and will spend the next few decades analysing them even further. It is perhaps only a matter of time that Kepler 186f will not be the only destination for us to aim towards. Spacecraft on such missions wouldn't be running interstellar errands to seek resources and materials, but in fact be like the sea faring ships of early settlers on Earth who walked across the land until they came to the sea - and then they crossed it, blind to what lay ahead of them.

A generational ship of such ambition would be a massive enterprise, built with the view of sustaining the long term survival of human beings. Such a ship will need to not only have the fastest propulsion possible but also be more efficient in managing its resources than human beings have ever been. The required technologies would need perfect harmony within the created ecosystem, a world capable of growing food from waste and processing waste to food to maximise energy efficiency. In such an interstellar ark the access to new resources would be near impossible as most of space is empty. The populations of such a craft would rely on

their biospheres for farming, enclosed ecosystems for sustenance and engineering systems that exist autonomously. They would need to co-operate, contributing to the survival of not only each other but the next generations of travellers as well. Their descendents would be educated and trained in the skills necessary to sustain such a ship and every crew member would serve an integral purpose to the journey. Machines like 3D printers would be used to manufacture all that is required - to repair, build and upgrade technology as the years went on - innovation wouldn't be a consumerist pursuit, discovery would be a matter of basic survival.

Second Star to the Right

Most of what we hope for in the designs and technology of interstellar ships may be seen as frivolous but it is in these instances of whimsy that insight shines brightest; like Einstein dreaming of riding a beam of light we are driven by the joy of discovery and not the fear of ignorance - it is this fearless drive towards discovery that will pave the way towards the next large leap into the cosmos.

As a species we must recognise that technology is the instrument of our will and science the dogma of our faith. The way we imagine, experiment and design our dreams defines what we can expect our future to be like. In that journey, be it towards the heavens in search of new homes or into the microcosm of our own selves we shall never truly discover unless we imagine. 